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An Overview of Chestnut Production in Turkey

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Abstract

This paper introduces chestnut production methods commonly practiced in Turkey, including germplasm, horticultural care, weed management, fertilization, orchard floor, insect and disease management, harvest, storage and marketing. Recommendations also are provided for improving chestnut production.

CHESTNUT PRODUCTION AND GERMPLASM

Turkey is third in the world for chestnut (*Castanea* spp.) production (59,171 t) behind China and South Korea (FAOSTAT, 2012). Chestnut production decreased from 1980 to 2005 primarily due to ink disease (*Phytophthora* spp.) discovered for the first time between 1940-1950 (Erdem, 1951) and chestnut blight (*Cryphonectria parasitica*) discovered in the 1960s (Akdogan and Erkam, 1968). Since 2005, control of chestnut blight by hypovirulence and the establishment of new orchards have led to an increase in chestnut production in Turkey. The main chestnut growing areas in Turkey are the Black Sea, Marmara and Aegean regions.

The Marmara region has the oldest and richest chestnut culture in the country. Many chestnut cultivars have been grown in this region for centuries. Eleven chestnut cultivars have been registered in the Marmara region (TTSM, 2012). Among these, 'Osmanoglu,' 'Sarıaslama,' 'Firdola' and 'Hacıomer' are the most frequently found cultivars in orchards. However, cultivars 'Marigoule' and 'Maraval,' hybrids of European (*Castanea sativa*) and Japanese chestnut (*C. crenata*), recently have been introduced in the region. The cultivar 'Marigoule' has become a preferred choice because it ripens early and is more tolerant to chestnut blight and drought.

In the Aegean region, grafted chestnuts are preferred. Ozkarakas et al. (1995) selected twenty-three chestnut genotypes for the Aegean Region. Ertan et al. (2007) carried out a new selection study in the Nazilli district and five chestnut genotypes were selected in this study for further evaluation. The most common cultivar in the region is 'Isıklar' (Sekerci). This cultivar is suitable for both roasting and processing into chestnut candy. Aydın province, located in the Aegean region, is the largest producer of chestnuts (34 %) in Turkey (TUIK, 2012).

Seedling trees are generally used for chestnut production in the Black Sea region, where several selection studies have been conducted (Ayfer et al., 1982; Serdar, 1999, 2002; Serdar and Soylu, 1999). An unregistered selection, 'Salipazari' (554-14 chestnut genotype) is a common local chestnut cultivar in the Salipazari and Terme districts of Samsun province. It was not officially registered because of its susceptibility to chestnut blight. However, five European chestnut cultivars were registered by Ondokuz Mayis University in 2009 (Serdar and Macit, 2010). Among these, cultivars 'Erfelek' and 'Unal' are becoming common in the Erfelek district of Sinop province. Another unregistered selection, 'Altinay' (SA 5-1 genotype), is a local cultivar grown in the Erfelek district of Sinop province (Serdar, 1999). However, it was never registered due to questions regarding its susceptibility to graft incompatibility as well as chestnut blight. Cultivar 'Marigoule,' has been tested since 2000 and registered for certification in young plant production (Serdar and Macit, 2010). Japanese chestnuts and additional hybrids of European and Japanese chestnut and some complex hybrids of different chestnut species have been tested at Ondokuz Mayis University and the Black Sea Agricultural Research Station since 2006 (Serdar and Macit, 2010).

CARING ACTIVITIES, HARVEST, STORAGE AND MARKETING

Few chestnut growers prune their chestnut trees regularly. Growers that do prune their trees remove limbs with symptoms of disease or pests. Few growers apply manure, fertilize, care for the orchard floor or irrigate chestnut trees since chestnut trees are generally found on hills and mountain slopes. However, most growers do attempt to manage weeds before harvesting the chestnuts. Chestnuts are harvested by pole or collection from the ground. Most of the growers don't store chestnuts; they generally collect chestnuts from the ground and market them immediately.

Chestnuts are usually stored in open areas as bur stacks and covered with ferns or other plant materials; these are termed "Gömü" or "Yığın". In these cases, growers store their crops and then sell them based on market demand. Ertan and Seferoğlu (2003) stated that total sugar and protein ratios of chestnuts decrease and starch, total carbohydrate and fat ratios increase after using this storage method. Chestnuts are generally consumed after roasting. Only a few growers store the chestnut in cold rooms. For chestnut candy production, companies generally store chestnuts in freezers as peeled kernels. Growers generally sell chestnuts directly to merchants or customers. Only a few growers sell chestnuts to cooperatives or chestnut candy companies. However, the chestnut processing sector has continued to develop into an important market for chestnuts in Turkey. There are thirty-three chestnut processing companies in Turkey; the original companies were located in the province of Bursa in the Marmara Region. Approximately 12,000 t of processed chestnuts are exported; the major products are candied chestnut, chestnut with chocolate bars, chestnut puree, chestnut flour, peeled chestnuts, chestnut snacks and chestnut cake.

DISEASES AND PESTS

In the past few years, the frequency of root rot disease affecting chestnut has increased in Turkey while chestnut blight, the most common disease on chestnut in Turkey, has decreased. Specifically, in the Marmara and Black Sea Regions, non-lethal blight infections on chestnut trees were determined to contain hypovirulent strains (Akilli et al., 2009a, b; 2011; Celiker and Onogur, 2009; Doken, 2009; Erincik et al., 2008). Before the biological control of chestnut blight was common, growers cut infected

branches or trees to control chestnut blight; however, some growers disinfected chestnut bark with applications of copper or lime.

Chestnut weevils are now the most important pest in Turkey. Growers attempt to eliminate infested nuts. However, some growers attempt to manage these pests with pheromones, bait traps or chemical control. Gall wasp (*Dryocosmus kuriphilus*) has not yet been discovered in Turkey.

RECOMMENDATIONS

Chestnut blight and root diseases have kept the nursery sector of the chestnut industry in Turkey from developing to its fullest extent. Cultivars with increased resistance to chestnut blight should be encouraged. Seedlings or clonal plants of European × Japanese hybrids should be used as rootstock in order to control root diseases. Establishing chestnut orchards in suitable forest areas and fields should be encouraged. Biological control of chestnut blight should be used commonly by growers. Research is needed in an effort to better manage chestnut weevil and nut rots. Grower cooperatives should be established in order to develop growing and marketing of chestnut. Storing chestnuts in cold rooms should be encouraged and suitable storage conditions and suitable packaging materials should be selected (Bilgener and Serdar, 1997; Jobling, 2000; Koyuncu et al., 2003; Yildizeli, 2003).

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